

Schedule

NRES 746

Fall 2025

Schedule

Note: this schedule is subject to change. Please check for updates frequently!

Week	Lecture 1	Lab	Lecture 2	Material Covered	Readings
Aug. 25	Course Introduction	Lab #1: Custom algorithms in R	Custom algorithms	Review syllabus, programming algorithms in R	Clark Ch. 1; Touchon and McCoy 2016; Bolker Ch 1
Sept. 1	No class (labor day)	Lab #1: Custom algorithms in R	Custom algorithms	Basic concepts in statistics, probability	Bolker ch. 4
Sept. 8	Probability	Lab #2: "Virtual Ecologist"	Probability	Probability, mechanistic models, stochastic models	Bolker Chs. 2, 3
Sept. 15	Data Generating Models	Lab #2: "Virtual Ecologist"	Data Generating Models	Power analysis, goodness-of-fit testing, likelihood	Bolker Ch. 5; Hobbs and Hilborn 2006 (optional)
Sept. 22	Likelihood	Lab #3: Maximum likelihood	Likelihood	Likelihood theory, maximum likelihood	Bolker Ch. 6
Sept. 29	Likelihood	Lab #3: Maximum likelihood	Optimization	Maximum likelihood inference, optimization algorithms	Bolker Chs. 6, 7
Oct. 6	Optimization	Group meetings: final projects	Bayesian inference	General introduction to Bayesian theory and application	Bolker Ch. 7 (Bayesian section); Ellison 2004
Oct. 13	Bayesian inference	Lab #4: Bayesian model fitting	Markov Chain Monte Carlo (MCMC)	Markov-Chain Monte Carlo for Bayesian inference	Bolker Ch. 7 (Bayesian section);
Oct. 20	Model selection and feature elimination	Lab #4: Bayesian model fitting	Model selection re: Bayesian inference	Model selection using information theoretic criteria	Bolker Ch. 6 (AIC section)
Oct. 27	Model performance evaluation	Student project progress reports and discussion	Model performance evaluation	Bias-variance tradeoff, cross-validation, assessing predictive accuracy	

Week	Lecture 1	Lab	Lecture 2	Material Covered	Readings
Nov. 3	Multilevel models	Demo/activity: multilevel models	Multilevel models	Multilevel models, “Random effects”, non-independence	
Nov. 10	Bayesian multilevel models	Group meetings: final projects	Bayesian multilevel models	Bayesian multilevel (hierarchical) models	
Nov. 17	Nonlinear regression, and GAMs	Demo/activity: GAMs in ‘mgcv’	Nonlinear models and GAMs	Generalized additive models (GAMs)	
Nov. 24	Spatial models	Demo/activity: Spatial models in INLA	Time-series models	Spatial autocorrelation	
Dec. 1	Time-series models	Demo/activity: time-series model in ‘stan’	Multivariate statistical models	Temporal autocorrelation	
Dec. 8	Class wrap-up- where to go from here	Student project presentations!	No class (prep day)		
Dec. 15	NA (classes over-project write-ups due by midnight Dec. 15)				